

PRODUCT REVIEW AND RATING MANAGEMENT SYSTEM

REVIEW AND RATING MANAGEMENT

Introduction

The Review and Rating module is an important part of the Amazon E-Commerce Management System. It allows customers to provide feedback about the products they purchase. Customers can give a rating from 1 to 5 and write a review describing their experience with the product. This module helps the system store customer feedback, retrieve product reviews, calculate average product ratings, and identify highly rated products.

1.REVIEW TABLE:

Column Name	Data Type	Constraint	Description
REVIEW_ID	NUMBER	Primary Key	Unique review ID
PRODUCT_ID	NUMBER	Foreign Key	Product being reviewed
CUSTOMER_ID	NUMBER	Foreign Key	Customer who gives the review
ORDER_ID	NUMBER	Foreign Key	Related order
REVIEW_TEXT	VARCHAR2(500)	—	Customer feedback
REVIEW_DATE	DATE	Default SYSDATE	Date of review

2. RATING TABLE

Column Name	Data Type	Constraint	Description
RATING_ID	NUMBER	Primary Key	Unique rating ID
PRODUCT_ID	NUMBER	Foreign Key	Rated product
CUSTOMER_ID	NUMBER	Foreign Key	Customer who gives the rating
RATING	NUMBER(1)	CHECK (1–5)	Rating given by customer
RATING_DATE	DATE	Default SYSDATE	Date of rating

3.CREATE REVIEW TABLE:

```

SQL> CREATE TABLE REVIEW (
  2     REVIEW_ID NUMBER PRIMARY KEY,
  3     CUSTOMER_ID NUMBER,
  4     PRODUCT_ID NUMBER,
  5     REVIEW_TEXT VARCHAR2(500),
  6     REVIEW_DATE DATE NOT NULL,
  7     CONSTRAINT FK_REV_CUST
  8         FOREIGN KEY (CUSTOMER_ID) REFERENCES CUSTOMER(ID),
  9     CONSTRAINT FK_REV_PROD
 10         FOREIGN KEY (PRODUCT_ID) REFERENCES PRODUCT(PRODUCT_ID)
 11 );

```

4.CREATE RATING TABLE:

```

SQL> CREATE TABLE RATING (
  2     RATING_ID NUMBER PRIMARY KEY,
  3     CUSTOMER_ID NUMBER,
  4     PRODUCT_ID NUMBER,
  5     RATING NUMBER CHECK (RATING BETWEEN 1 AND 5),
  6     RATING_DATE DATE NOT NULL,
  7     CONSTRAINT FK_RAT_CUST
  8         FOREIGN KEY (CUSTOMER_ID) REFERENCES CUSTOMER(ID),
  9     CONSTRAINT FK_RAT_PROD
 10         FOREIGN KEY (PRODUCT_ID) REFERENCES PRODUCT(PRODUCT_ID)
 11 );

```

Table created.

5.INSERING VALUES:

```

SQL> INSERT INTO REVIEW
  2 VALUES (1, 101, 102, 'Good product and quality is excellent', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO REVIEW
  2 VALUES (2, 102, 105, 'Product is useful and worth the price', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO REVIEW
  2 VALUES (3, 103, 108, 'Very satisfied with the product', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO REVIEW
  2 VALUES (4, 101, 105, 'Average product but can be improved', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO REVIEW
  2 VALUES (5, 102, 102, 'Good quality product', SYSDATE);

1 row created.

```

6.RETRIEVE PRODUCT REVIEW DETAILS:

```
SQL> SELECT
  2     R.REVIEW_ID,
  3     R.CUSTOMER_ID,
  4     R.PRODUCT_ID,
  5     P.PRODUCT_NAME,
  6     R.REVIEW_TEXT,
  7     R.REVIEW_DATE
  8 FROM REVIEW R
  9 JOIN PRODUCT P
 10 ON R.PRODUCT_ID = P.PRODUCT_ID;
```

7.CALCULATE AVERAGE PRODUCT RATINGS:

```
SQL> SELECT
  2     PRODUCT_ID,
  3     ROUND(AVG(RATING), 2) AS AVERAGE_RATING
  4 FROM RATING
  5 GROUP BY PRODUCT_ID;
```

PRODUCT_ID	AVERAGE_RATING
102	4.5
105	3.5
108	5

8.IDENTIFY HIGHLY RATED PRODUCTS:

```
SQL> SELECT
  2     P.PRODUCT_ID,
  3     P.PRODUCT_NAME,
  4     ROUND(AVG(R.RATING), 2) AS AVERAGE_RATING
  5 FROM RATING R
  6 JOIN PRODUCT P
  7 ON R.PRODUCT_ID = P.PRODUCT_ID
  8 GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9 HAVING AVG(R.RATING) >= 4
 10 ORDER BY AVERAGE_RATING DESC;
```

PRODUCT_ID

PRODUCT_NAME

AVERAGE_RATING

108
Laptop Bag 5

102
Wireless Headphones 4.5

PRODUCT_ID

PRODUCT_NAME

AVERAGE_RATING